

Valerio Valente

Data Scientist - AI Engineer

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Professional Summary

Data Scientist and AI Engineer with 3 years of experience shipping AI agents, RAG systems and MLOps pipelines for pharma, manufacturing and research clients. Currently at Aixelerate, where I own the AI layer of a research platform - building agent-driven analysis tools, explainable statistical components and the product surfaces around them. Prior work: regulatory-compliant RAG (top 5 FDA challenge), tabular synthetic data, reproducible Azure ML pipelines.

Skills

- **AI Agents & LLM Systems** - RAG, multi-agent systems, MCP servers, explainable AI outputs. OpenAI, HuggingFace, LangChain, prompt engineering, evals.
- **ML & Statistical Methods** - Supervised/unsupervised learning, tabular synthetic data, statistical analysis components for research workflows. PyTorch, Scikit-Learn, Pandas.
- **MLOps & Cloud** - Reproducible ML pipelines, experiment tracking, scalable inference. Azure ML, AWS, CI/CD, containerization.
- **Product & Software Engineering** - Production Python (FastAPI), React/TypeScript frontend, git, testing, CI quality gates.

Work Experience

Data Scientist - AI & Product (Aixelerate s.r.l.)

April 2026 - current

- Built and shipped a suite of **statistical-analysis** components for clinical datasets, running end-to-end from input validation to user-facing outputs.
- Own the evolution of the AI layer toward a production multi-agent architecture **moving the system from development to production-grade stage**.
- Authored **Claude Code skills** standardizing quality and cutting repetitive scaffolding work.
- Contributed across the product stack with a strong UX focus and pushing usability improvements informed by researcher interviews.

Data Scientist (InSilicoTrials Technologies SpA)

April 2024 – March 2026

- Designed and maintained the company chatbot, a multi agent system that included a **Model Context Protocol (MCP)** server, improving users' knowledge retrieval and support workflows.
- **Contributed to the open-source AI project mcpadapt** on GitHub, strengthening knowledge of advanced Python patterns and aligning with industry standards.
- Delivered a complete workflow for **tabular synthetic data generation**, employing SOTA tabular foundation model.
- Placed in the **top 5 in the FDA challenge** by delivering a **regulatory compliant RAG model**, complete with **custom evaluation benchmarks** that improved robustness and traceability.
- Built and deployed reproducible **ML pipelines on Azure**, supporting research teams through standardized experiment tracking and scalable inference workflows.

Junior Data Scientist (Miraitec srl)

September 2023 - April 2024

- Built a **RAG retrieval agent on AWS** that automated document analysis and reduced manual processing time for manufacturing clients.
- Surfaced operational inefficiencies through pattern recognition (7% production-performance improvement) and BI dashboards (Apache Superset + SQL).

Education

Master of Science in Engineering in Computer Science (Sapienza University)

January 2023

Major: Computer Science | Minor: Data Science & Machine Learning

- Thesis: *NLP for Integrity on Social Platforms* (collaboration with **ex-Meta researcher**).
- Achieved +15% accuracy improvement using meta-learning techniques over traditional approaches.

Bachelor of Science in Software Engineering (Roma Tre University)

December 2019

Major: Engineering | Minor: Computer Science

Achievements & Certifications

- Top 3.6% of HUAWEI Seeds For the Future (2022) - business plan focused on tech innovation, budget and investor engagement. Additional ML/MLOps/AI certifications on LinkedIn.